IRCA Exposure and Composition of Admissions

A descriptive exposure-gradient analysis, 1982–2000

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Executive Summary

This project builds a reproducible country–year panel (19 Latin American origins, 1982–2000) to document whether IRCA legalization intensity (1989–1992) is associated with changes in the recorded occupational composition of admissions.

What I find: The blue-collar "share decline" during the IRCA surge depends on how you define the denominator. Using known occupations only, higher exposure is linked to a sizeable drop in the blue-collar share. When I include "no occupation" in the denominator, the association largely disappears—suggesting the KPI is driven by reporting/denominator shifts during surge years.

How to read the estimates: This is a descriptive exposure-gradient analysis, not a causal design. Event-study timing checks show pre-period differences, so I use the regression results as summary associations and focus on robustness plus numerator/denominator decomposition.

Reproducibility: All tables/figures are generated from the raw Excel inputs via run.R (panel build → QA → models → exports).

Portfolio Exhibits

If you skim: start with Figure 1 → Table 1 → Figure 2, then use Figures 3–4 as diagnostics.

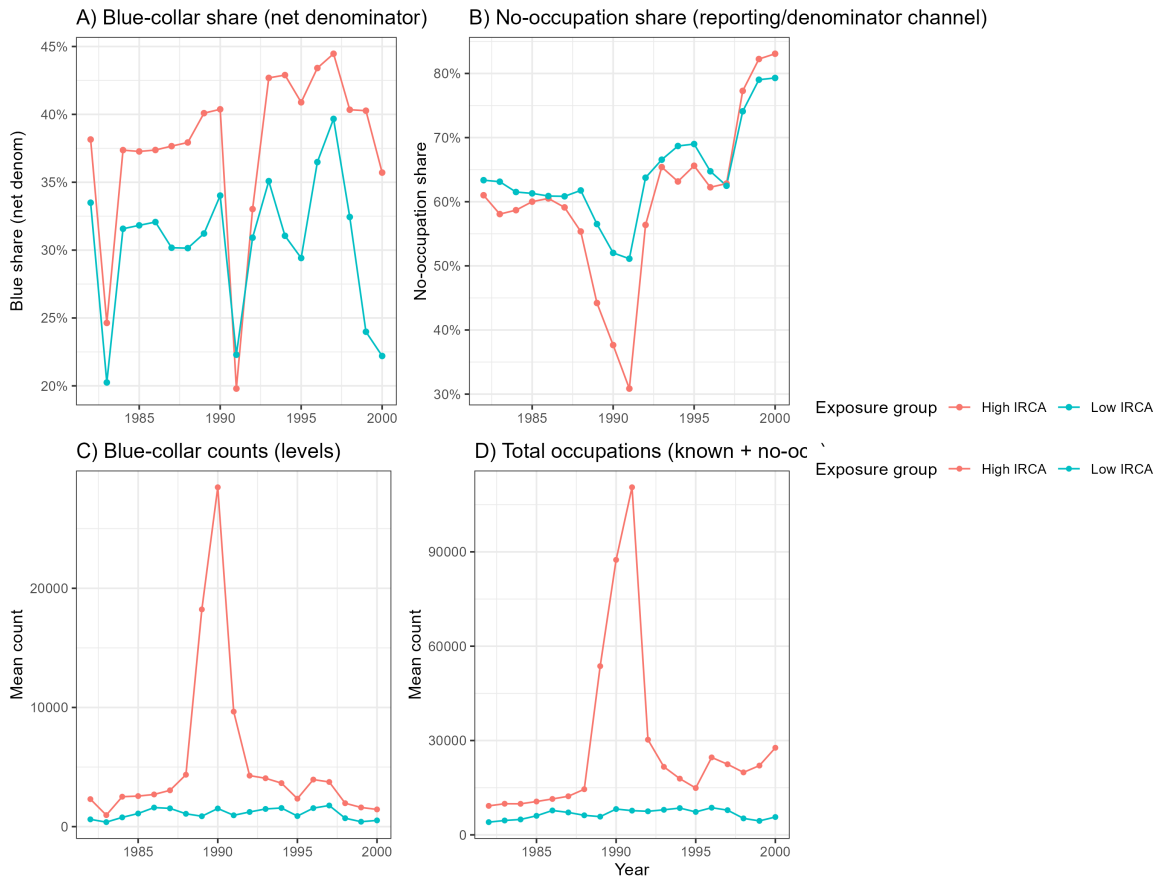

Figure 1: Shares, no-occupation share, and counts by IRCA exposure group.

Research Question and Data

The input spreadsheets in data_raw/ were compiled by the author from publicly available INS Statistical Yearbooks. The report uses only the series required for this analysis (occupation, class of admission, gender/age, adjustments vs new arrivals). See Sources section for publication references.

Question: Do origin countries with higher IRCA legalization intensity display larger changes in the occupational composition of admissions during 1989–1992?

Exposure (country-level):

$$\text{Exposure} = \frac{\text{IRCA legalizations (1989–1992)}}{\text{Pre-IRCA admissions (1982–1985)}}.$$

Data sources: INS Statistical Yearbooks / OHSS tables on occupation, class of admission, gender/age, and adjustments vs new arrivals.

Design: descriptive exposure-gradient panel regressions with country and year fixed effects; clustered SE at the origin-country level.

Main Result: Denominator Sensitivity

The main portfolio result is that the blue-share association depends strongly on whether "no occupation" is excluded (net denominator) or included (full denominator). To avoid sample-driven artifacts, the headline comparison uses the same set of country–years for both outcomes.

Table 1: Blue share with Net vs Full denominator (same sample).

	Blue share (Net denom; same sample)	Blue share (Full denom; same sample)
IRCA intensity × Post	−0.036***	−0.005
	(0.007)	(0.005)
Adjustment share	0.169*	0.127**
	(0.085)	(0.059)
Num.Obs.	351	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Same-sample comparison: rows require both blue_share and blue_fullden observed. Portfolio takeaway: denominator sensitivity suggests reporting/missingness mechanics are central.

Given the timing diagnostics (Appendix E), interpret this coefficient as a descriptive summary of the exposure gradient—not a causal estimate. In practice: the headline shift is largely explained by changes in occupation coverage (how many records have usable occupation codes), so the dashboard should always pair the net share with the full-denominator share and the no-occupation share.

Practical implication: during large legalization/processing surges, composition metrics based on "known occupations only" can move sharply even when the underlying category counts rise — because denominators (including missing/unknown occupation mass) change. For example, if more records are processed without occupation coding during surge years, the 'known-occupation' denominator changes even if blue-collar counts rise.

Mechanism: Numerator vs Denominator Growth

To understand why a share can fall even when counts rise, I decompose the association into numerator and denominator components (levels). The pattern is consistent with denominator growth outpacing numerator growth, rather than a collapse in blue-collar admissions.

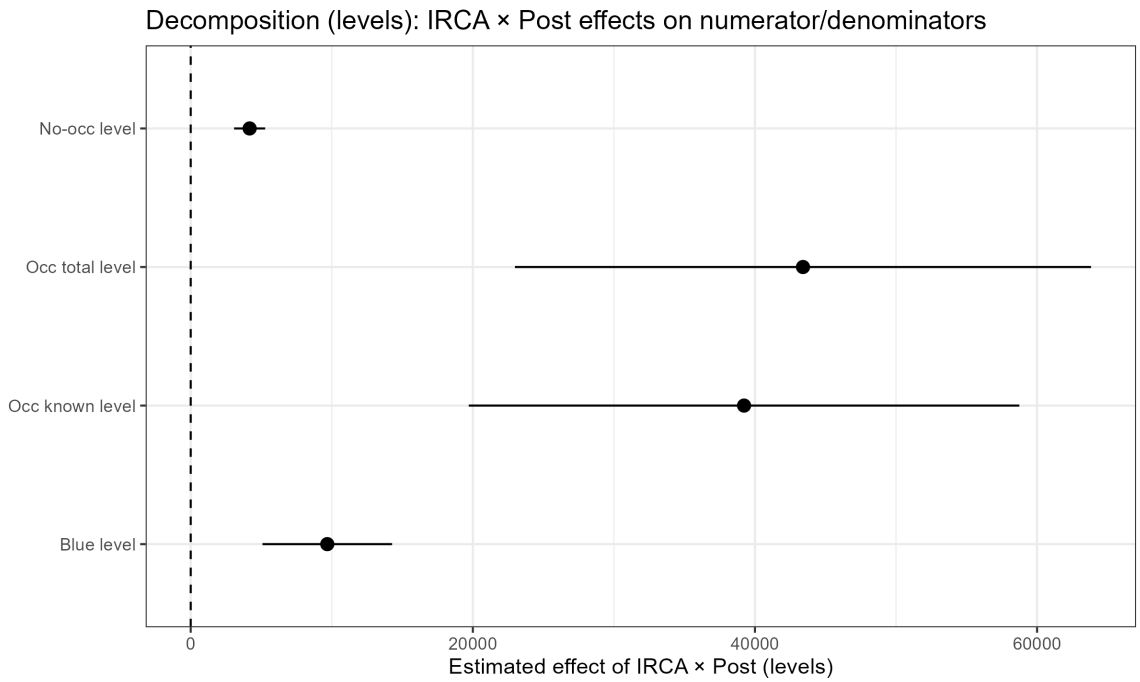

Figure 2: IRCA × Post associations on numerator (blue counts) and denominators (levels).

Denominator Channel Evidence: "No Occupation"

A key diagnostic for denominator mechanics is whether the "no occupation" category shifts differentially by exposure during the surge.

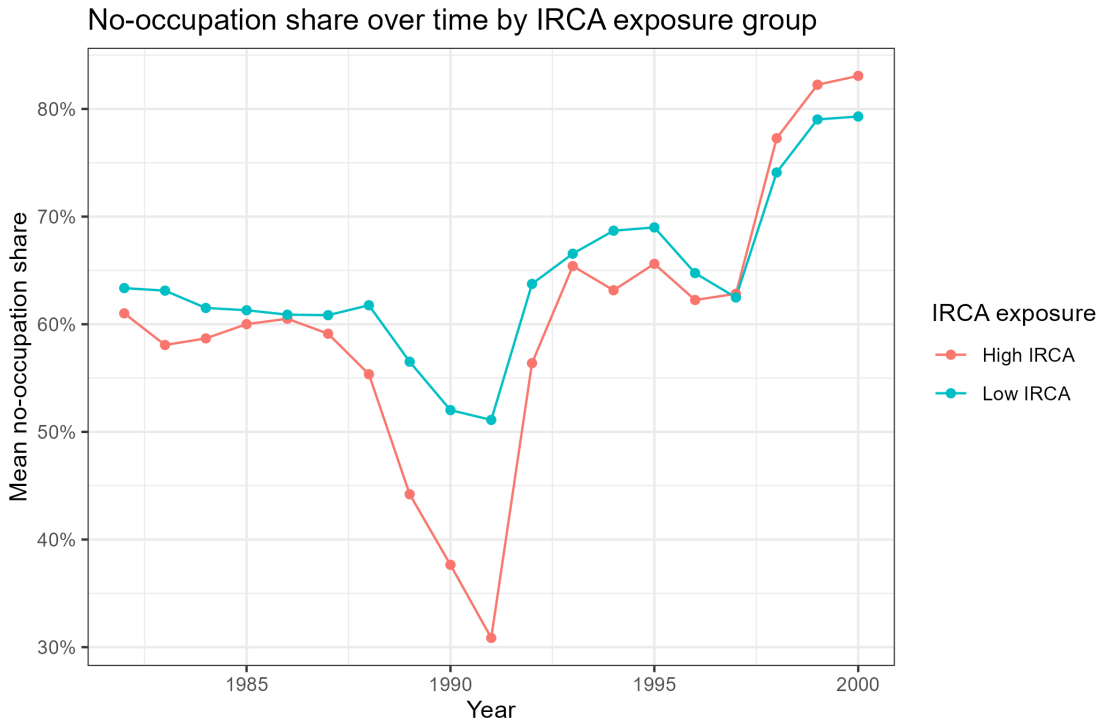

Figure 3: No-occupation share over time by IRCA exposure group.

Robustness

The negative net-denominator association is not driven by a single origin country.

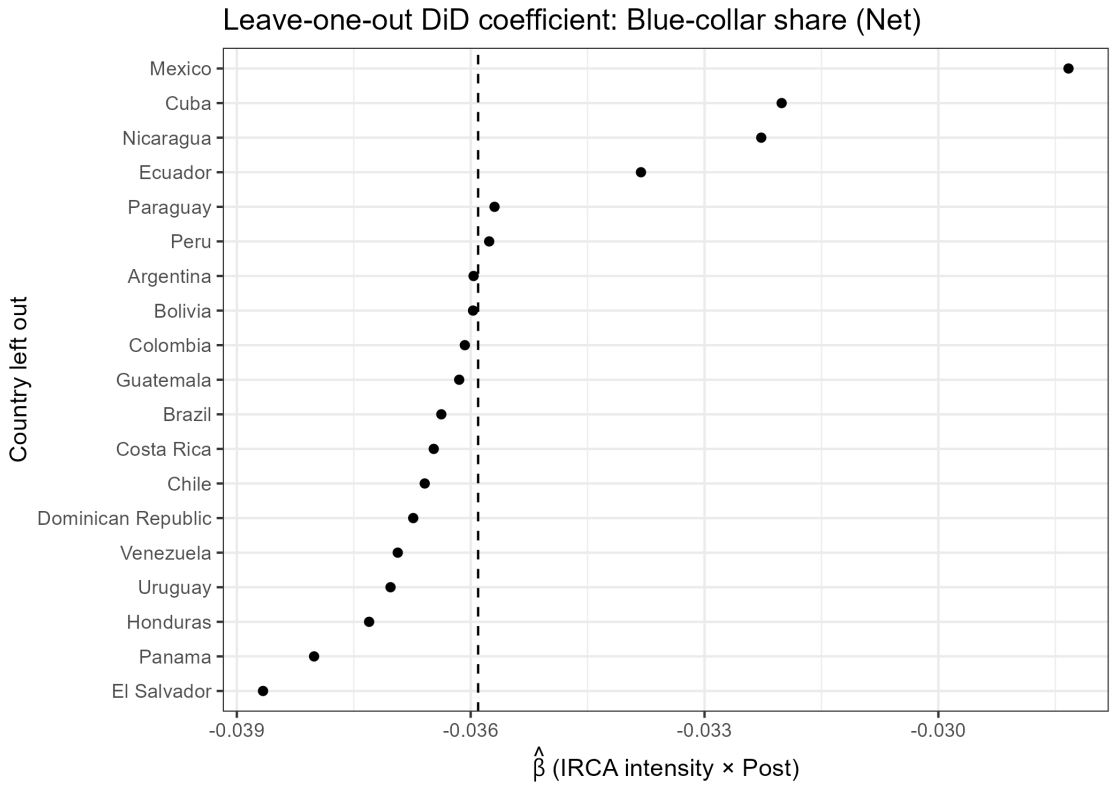

Figure 4: Leave-one-country-out influence on the IRCA × Post coefficient (net denominator).

Small-cluster inference note: with 19 clusters, conventional clustered SE can be fragile. I therefore include a wild cluster bootstrap p-value for the baseline specification (Appendix contains full robustness tables).

Table 2: Small-cluster inference add-on: Wild cluster bootstrap test for IRCA × Post (baseline net specification).

Model	Parameter	Bootstrap p-value	Bootstrap t-stat
lm with country+year FE (dummies)	IRCA intensity × Post	0.03361	-5.154

Wild bootstrap computed on an equivalent country+year dummy specification (same model, different implementation) due to package compatibility.

Magnitudes

Because the headline depends on denominator definition, magnitudes are reported for both constructions.

Table 3: Standardized effect sizes for net-denominator blue share (same-sample comparison).

Magnitude	Effect (pp / %)
Per 1 unit exposure	-3.59
Per 1 SD exposure	-7.61
Per IQR exposure	-2.46
As % of mean(Y): 1 SD	-22.22
As % of mean(Y): IQR	-7.18

Table 4: Standardized effect sizes for full-denominator blue share (same-sample comparison).

Magnitude	Effect (pp / %)
Per 1 unit exposure	-0.54
Per 1 SD exposure	-1.13
Per IQR exposure	-0.37
As % of mean(Y): 1 SD	-8.63
As % of mean(Y): IQR	-2.79

Stakeholder and Decision context (DA framing)

Intended stakeholder and decision use

Who this is for: an admissions reporting / operations analytics team (e.g., OHSS-style reporting or an internal USCIS analytics group).

Decision use: during surge windows (large legalization/processing events), this analysis supports:

- **Monitoring:** detect whether composition changes reflect real shifts vs. reporting/denominator changes.
- **Quality assurance:** flag periods when "no occupation" rises and occupation coding may be less consistent.
- **Capacity planning:** anticipate downstream staffing needs if denominator growth reflects processing/recording expansion.

Metrics specifications (dashboard KPIs)

All shares are in $[0, 1]$ and should be displayed as percentage points in dashboards. For each origin country c and year t:

- **Blue share (net denominator)**

$$\text{BlueShare}_{ct}^{net} = \frac{\text{BlueCollarNonFarm}_{ct}}{\text{OccKnown}_{ct}}$$

  where

$$\text{OccKnown}_{ct} = \text{WhiteHigh}_{ct} + \text{WhiteSupport}_{ct} + \text{BlueCollarNonFarm}_{ct} + \text{Agriculture}_{ct} + \text{Services}_{ct}.$$

- **Blue share (full denominator)**

$$\text{BlueShare}_{ct}^{full} = \frac{\text{BlueCollarNonFarm}_{ct}}{\text{OccKnown}_{ct} + \text{NoOccupation}_{ct}}.$$

- **No-occupation share (full denominator)**

$$\text{NoOccShare}_{ct} = \frac{\text{NoOccupation}_{ct}}{\text{OccKnown}_{ct} + \text{NoOccupation}_{ct}}.$$

In the pipeline (run.R), these correspond to blue_share (net), blue_fullden (full), and noocc_share.

Why report all three together:

BlueShare^{net} can move sharply if the **denominator changes** (e.g., classification shifts into "no occupation" or changes in reporting coverage). Reporting BlueShare^{full} and NoOccShare alongside the net share helps distinguish **true compositional change** from **measurement/reporting mechanics**.

How to Run

1. Put the raw Excel inputs in data_raw/.
2. Open `irca-composition.Rproj`
3. Run `Rscript run.R` (or Source `run.R`)
4. Render `paper.Rmd`
5. Outputs are written to:

- outputs/tables/, outputs/figures/, outputs/logs/, and data_clean/panel.rds

Sources

- U.S. Immigration and Naturalization Service (INS). *Statistical Yearbook of the Immigration and Naturalization Service*, various years (occupation and admissions tables).

Appendix

Appendix A: Data QA (missingness and coverage)

Table 5: Missingness by variable at the country–year level.

Variable	Obs.	Missing	Total	Share missing (%)
Male share	133	228	361	63.2
Family share	352	9	361	2.5
Employment share	352	9	361	2.5
Humanitarian share	352	9	361	2.5
Other share	352	9	361	2.5
Adjustment share	352	9	361	2.5
Blue share	353	8	361	2.2
Agriculture share	353	8	361	2.2
Manual share	353	8	361	2.2
Services share	353	8	361	2.2
White share	353	8	361	2.2
No-occupation share	353	8	361	2.2

Table 6: Coverage by origin country for key variables (share of years observed).

Country	Years total	Years covered (key vars)	Coverage ratio
Argentina	19	19	1.000
Bolivia	19	19	1.000
Brazil	19	19	1.000
Chile	19	19	1.000
Colombia	19	19	1.000
Costa Rica	19	19	1.000
Cuba	19	19	1.000
Dominican Republic	19	19	1.000
Ecuador	19	19	1.000
El Salvador	19	19	1.000
Guatemala	19	19	1.000
Honduras	19	19	1.000
Mexico	19	19	1.000
Nicaragua	19	19	1.000
Panama	19	19	1.000
Peru	19	19	1.000
Venezuela	19	19	1.000
Uruguay	19	15	0.789
Paraguay	19	13	0.684

Appendix B: Baseline regression panels

Table 7: Baseline TWFE (Total): occupation shares (Blue–White).

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.025***	0.029***	0.004	−0.007**	0.004
	(0.005)	(0.004)	(0.005)	(0.003)	(0.005)
Num.Obs.	353	353	353	353	353
Mean of Y (share units)	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 8: Baseline TWFE (Total): no-occupation share and class-of-admission shares.

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.035***	−0.052***	−0.021*	−0.004***	−0.002
	(0.007)	(0.004)	(0.011)	(0.001)	(0.001)
Num.Obs.	353	352	352	352	352
Mean of Y (share units)	0.620	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 9: Baseline TWFE (Net): occupation shares (Blue–White), controlling for adjustment share.

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.036***	0.029***	−0.007	−0.006*	0.013*
	(0.007)	(0.004)	(0.006)	(0.003)	(0.006)
Adjustment share	0.169*	−0.006	0.164*	−0.019	−0.144
	(0.085)	(0.027)	(0.082)	(0.032)	(0.093)
Num.Obs.	351	351	351	351	351
Mean of Y (share units)	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 10: Baseline TWFE (Net): no-occupation share and class-of-admission shares, controlling for adjustment share.

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.025***	−0.027**	−0.008	−0.016**	−0.009
	(0.005)	(0.011)	(0.011)	(0.006)	(0.009)
Adjustment share	−0.149*	−0.400**	−0.206	0.192**	0.108
	(0.081)	(0.152)	(0.128)	(0.091)	(0.138)
Num.Obs.	351	352	352	352	352
Mean of Y (share units)	0.619	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Appendix C: Additional robustness tables

Table 11: Specification comparison (blue share, net): baseline vs trends, weights, and balanced panel.

	Baseline Net	Country trends	WLS admissions	Balanced Net
IRCA × Post	−0.036***	−0.036***	−0.036***	−0.037***
	(0.007)	(0.008)	(0.003)	(0.007)
Adjustment share	0.169*	0.160	0.150**	0.190*
	(0.085)	(0.099)	(0.069)	(0.090)
Num.Obs.	351	351	351	323

* p <0.1, ** p <0.05, *** p <0.01
Comparison of key robustness specifications for blue-collar share.

Table 12: Robustness to origin-specific linear trends (blue share).

	Baseline Total	Baseline Net	Trends Total	Trends Net
IRCA intensity × Post	−0.025***	−0.036***	−0.025***	−0.036***
	(0.005)	(0.007)	(0.005)	(0.008)
Adjustment share		0.169*		0.160
		(0.085)		(0.099)
Num.Obs.	353	351	353	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Trends specs include origin-specific linear trends (country[t]).

Table 13: Additional sensitivity checks (blue share, net): drop high-missing years, drop incomplete-coverage countries, and restrict window.

	Baseline Net	Drop 1995–1997	Drop UY+PY	Window 1982–1995
IRCA × Post	−0.036***	−0.042***	−0.037***	−0.037***
	(0.007)	(0.008)	(0.007)	(0.008)
Adjustment share	0.169*	0.224**	0.190*	0.182*
	(0.085)	(0.092)	(0.090)	(0.092)
Num.Obs.	351	300	323	260

* p <0.1, ** p <0.05, *** p <0.01
All models: Country & Year FE; SE clustered by country. Outcome: blue_share.

Table 14: Weighted specifications (blue share, net): admissions weights and occupation-count weights.

	Net (unweighted)	Net (w: admissions)	Net (w: known occ.)
IRCA intensity × Post	−0.036***	−0.036***	−0.034***
	(0.007)	(0.003)	(0.004)
Adjustment share	0.169*	0.150**	0.105
	(0.085)	(0.069)	(0.084)
Num.Obs.	351	351	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country.

Table 15: Balanced-panel restriction (blue share, net).

	Blue Net (balanced)
IRCA × Post	−0.037***
	(0.007)
Adjustment share	0.190*
	(0.090)
Num.Obs.	323

* p <0.1, ** p <0.05, *** p <0.01
Balanced panel: keeps only countries observed in all years of the Net sample.

Table 16: Alternative exposure construction and Mexico exclusion (blue share, net).

	Blue Net (baseline)	Blue Net (alt denom)	Blue Net (–MX)	Blue Net (alt –MX)
IRCA intensity × Post	−0.036***		−0.028**	
	(0.007)		(0.013)	
IRCA intensity (alt) × Post		−0.009***		−0.007**
		(0.002)		(0.003)
Adjustment share	0.169*	0.169*	0.163	0.163
	(0.085)	(0.085)	(0.106)	(0.106)
Num.Obs.	351	351	332	332

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Outcome is share in [0,1]; coefficients are pp.

Appendix D: Decomposition tables + log plot

Table 17: Decomposition (levels): IRCA × Post associations on blue counts and denominators.

	Blue level	Occ known level	Occ total level	No-occ level
IRCA × Post	9682.575***	39 226.652***	43 404.916***	4178.264***
	(2342.817)	(9956.342)	(10 418.047)	(561.725)
Adjustment share	−973.413	−18 202.159	−204.179	17 997.980
	(5699.283)	(21 227.495)	(32 984.662)	(17 559.005)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Levels are counts.

Table 18: Decomposition (log(1+x)): proportional associations on blue counts and denominators.

	ln(1+Blue)	ln(1+Occ known)	ln(1+Occ total)	ln(1+No-occ)
IRCA × Post	0.098*	0.192***	0.131***	0.056
	(0.051)	(0.037)	(0.036)	(0.034)
Adjustment share	1.092	0.776	0.608	0.224
	(0.636)	(0.469)	(0.470)	(0.513)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Log specs use ln(1+x).

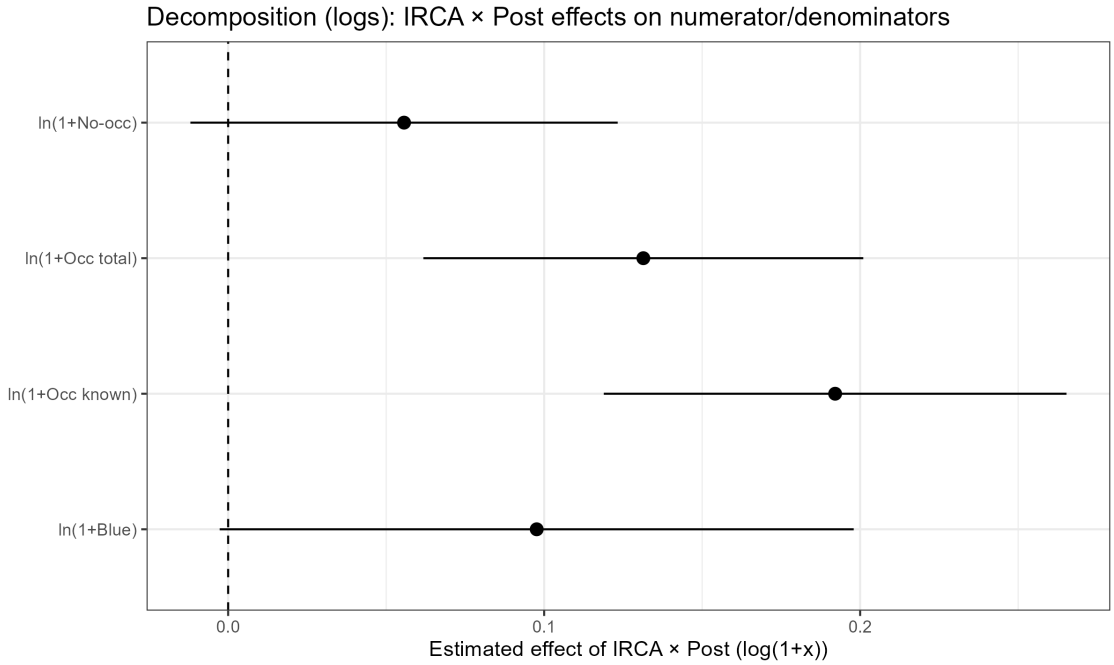

Figure 5: Decomposition in log(1+x): proportional associations on numerator and denominators.

Appendix E: Timing diagnostics (interpretation check)

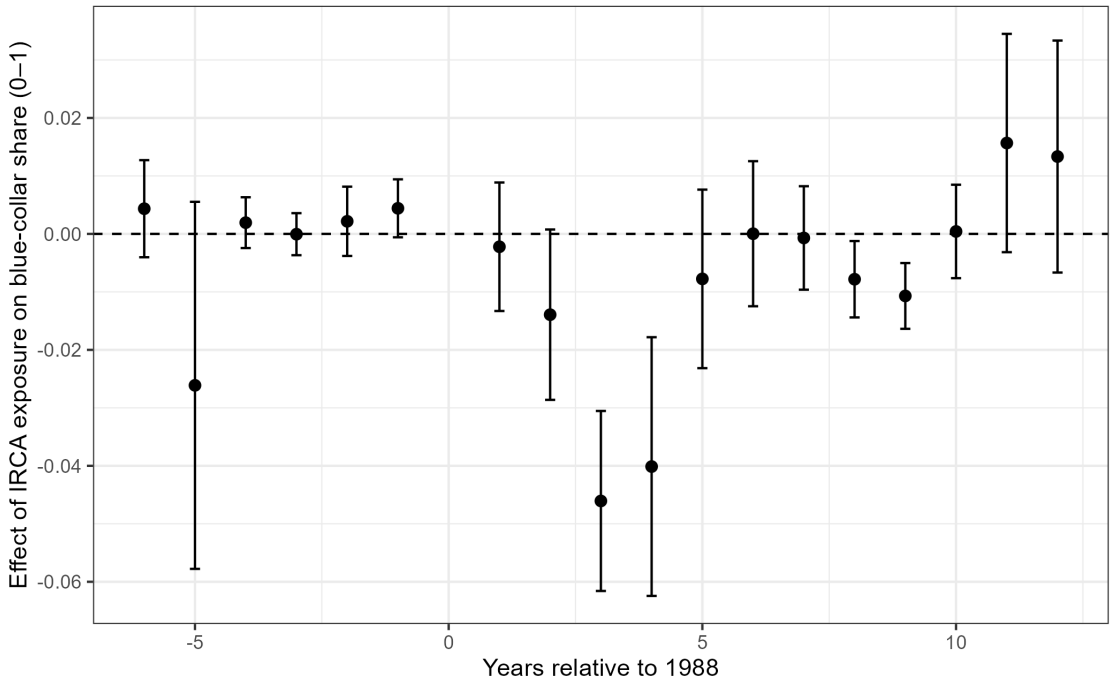

Figure 6: Event-study: exposure-by-year associations for blue-collar share (relative to 1988).

Table 19: Event-study pretrend diagnostic: Wald test of joint significance for pre-1988 coefficients.

Test	Statistic	df1	df2	p_value
Wald joint pre-trends (years < 1988)	13.54	6	298	0

Note: p-value is extremely small; it prints as 0 due to formatting/rounding.

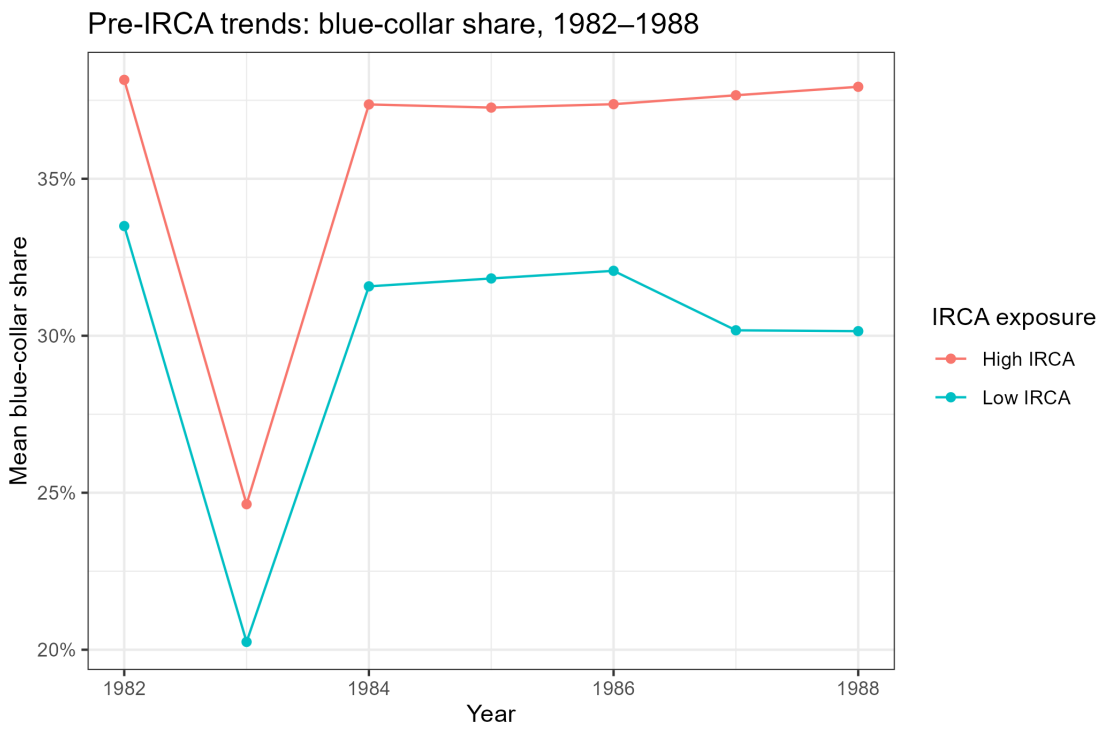

Figure 7: Pre-IRCA group means (1982–1988): blue-collar share by IRCA exposure group.

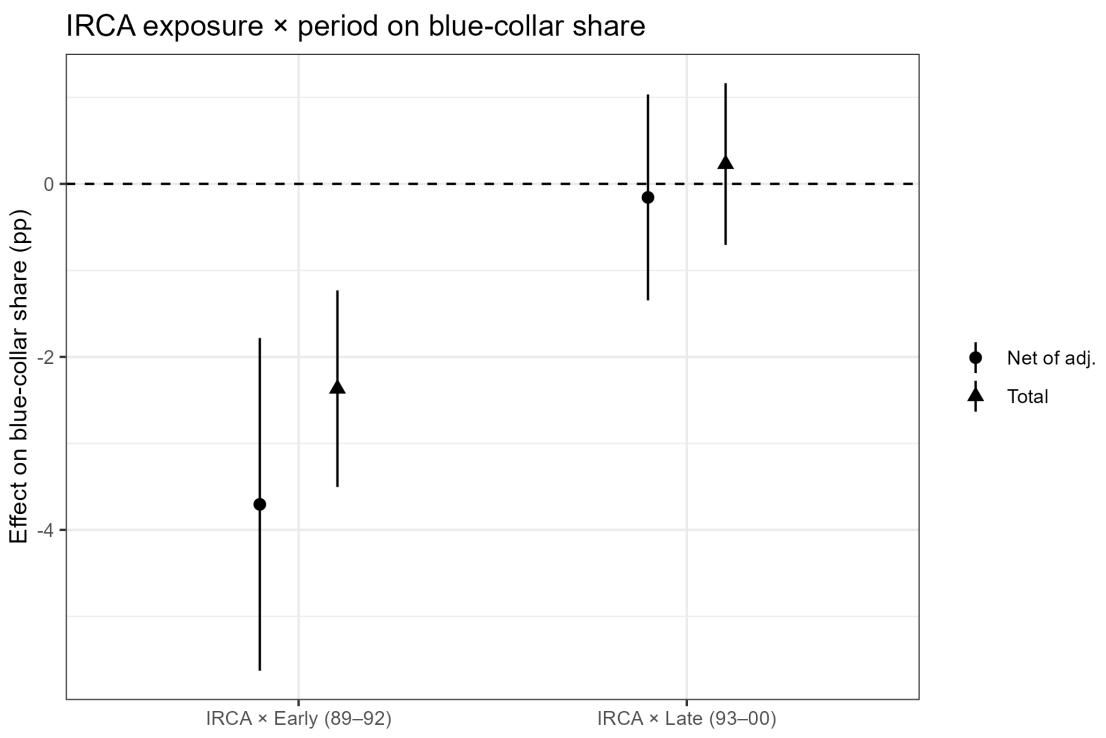

Figure 8: Early vs late post-period associations on blue-collar share (exposure-gradient).

Additional timing checks (linear pretrend regression and placebo fake-post tests) are available in logs/tables but omitted from the PDF for brevity.

Appendix F: Exploratory analyses

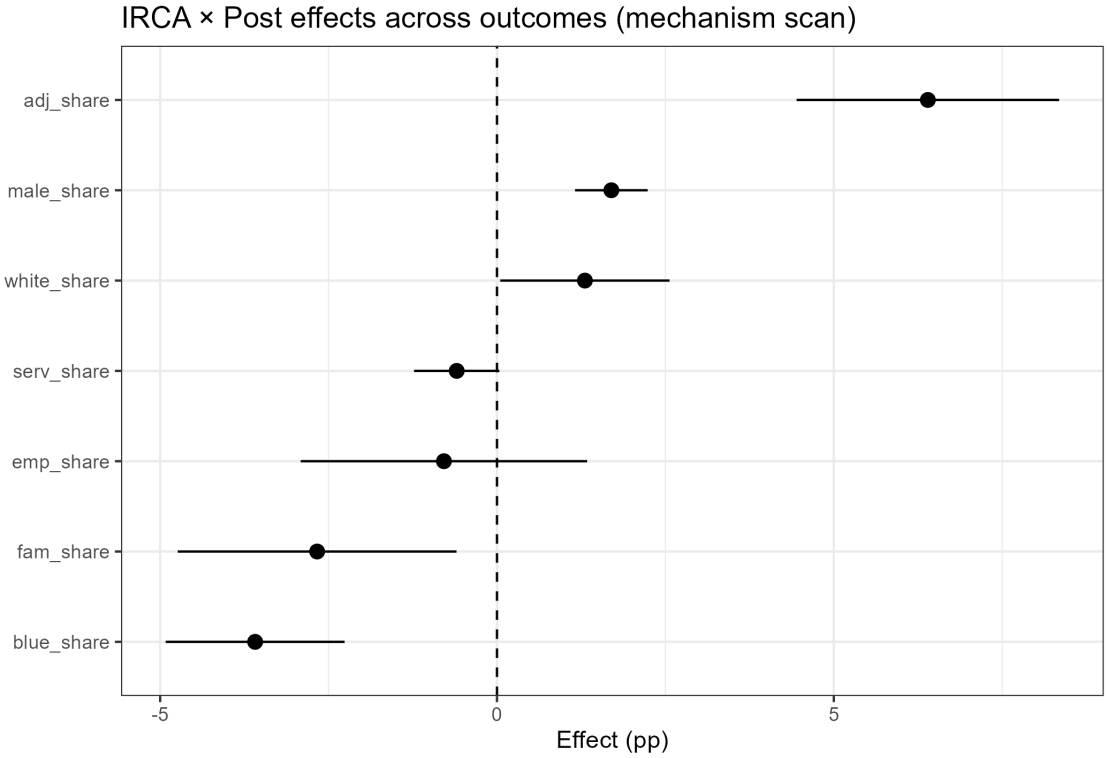

Figure 9: Exposure × Post associations across outcomes (exploratory).

Appendix G: QA artifact (residual sanity check)

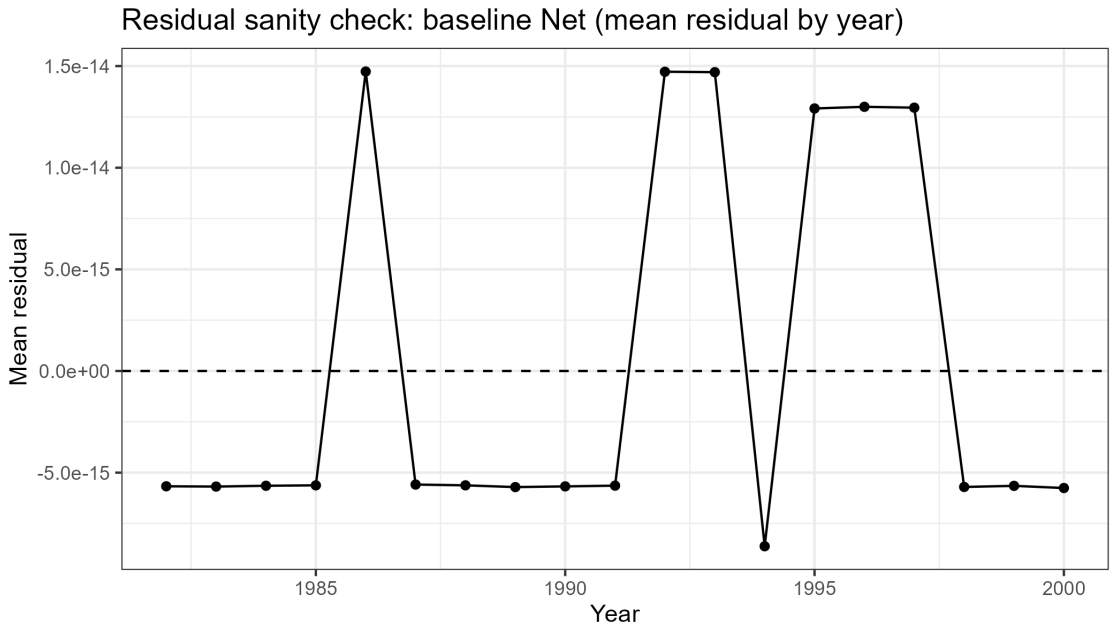

Figure 10: Mean residual by year (baseline net specification).